

Project 2: Exploratory Data Analysis & Interactive E-Commerce Dashboard Documentation

DecodeLabs Data Analytics Internship 2026

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Dataset Scope: 1,200 transactional records (Jan 2023 - Jun 2025)

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Executive Summary

This report outlines the research findings and software documentation for Project 2 of the DecodeLabs Data Analytics program. The purpose of this investigation was to perform a rigorous exploratory data analysis (EDA) on an e-commerce transactional dataset containing 1,200 records, and subsequently develop a premium interactive dashboard to visualize key performance metrics, temporal trends, operational risks, and statistical anomalies.

The analysis reveals a structurally sound revenue foundation, with total generated revenue reaching \$1.26 million. However, significant operational risks were identified: specifically, the high cancellation and return rate, which cumulatively accounts for 41.4% of all orders. The interactive dashboard addresses these business dimensions, enabling decision-makers to perform real-time queries and filter transaction data dynamically by time, product category, and risk profile.

1. Descriptive Statistical Overview

A descriptive statistical summary was conducted across the primary numerical dimensions of the dataset, including order quantity, item unit pricing, items per cart, and final transaction pricing. The metrics highlight a distinct right-skewness, particularly in transaction pricing, indicating the presence of high-value bulk purchases.

Metric	Quantity	Unit Price (\$)	Items In Cart	Total Price (\$)
Mean	2.95	356.41	5.49	1,053.97
Median	3.00	364.21	5.00	823.62
Std Dev	1.41	197.18	2.28	819.86
Minimum	1.00	11.39	1.00	11.39
Maximum	5.00	699.93	10.00	3,456.40

Key Statistical Observations:

- **Mean vs. Median Discrepancy:** The average order price (\$1,053.97) is significantly higher than the median order price (\$823.62). This right-skewed distribution indicates that high-value transactions disproportionately influence overall revenue projections.
- **Order Size Limits:** Quantities are strictly bounded between 1 and 5 items per transaction, whereas items in the cart range up to 10, indicating bulk packaging preferences among consumers.

2. Outlier Detection via Interquartile Range (IQR)

Using the Interquartile Range methodology on the Total Price variable, an upper boundary fence was established at \$3,330.41 (calculated as $Q3 + 1.5 * IQR$, where $Q1 = \$410.52$ and $Q3 = \$1,578.48$). Under this threshold, eight outlier records were detected. A granular audit of these records revealed a consistent profile: all records represent transactions with maximum quantity (5 units) and high unit prices ($> \$666$). These profiles represent bulk B2B purchases rather than clerical or system errors, and should be isolated for targeted enterprise account management.

Order ID	Product	Qty	Unit Price (\$)	Total Price (\$)
ORD200107	Printer	5	670.75	3,353.75
ORD200326	Laptop	5	670.48	3,352.40
ORD200328	Tablet	5	674.04	3,370.20
ORD200469	Chair	5	676.98	3,384.90
ORD200632	Laptop	5	678.16	3,390.80
ORD200789	Tablet	5	691.28	3,456.40
ORD201065	Printer	5	666.80	3,334.00
ORD201122	Monitor	5	678.19	3,390.95

3. System Architecture and Implementation

To allow stakeholders to dynamically explore these data insights, a premium single-page React web application was built using a modular component design. The application implements standard React state management hooks to support fluid interactions.

Key Interactive Components:

- **Reactive Temporal Filter:** A stateful selector that partitions data by calendar year, immediately updating descriptive stats, transaction frequencies, and timeline trends.
- **Sortable Metrics Grid:** A dynamic product table that sorts categories by revenue, order count, or average transaction value based on user headers selection.
- **Click-to-Inspect Charts:** Integrated timeline nodes in Recharts that show precise monthly performance overlays upon selection.

4. Strategic Business Recommendations

Based on the exploratory data analysis, the following structural adjustments are recommended for DecodeLabs operational workflows:

- **Address the 41.4% Fulfillment Risk:** With Cancelled (20.8%) and Returned (20.6%) orders representing a significant loss, a root-cause analysis is required. Specific attention should be paid to Monitor and Tablet lines, which exhibit the highest cancellation rates (43.6% and 43.0% respectively).
- **Capitalize on Instagram Channels:** Instagram generates 21.8% of total revenue (\$275,285), making it the top customer source. Marketing resources should be reallocated from lower-performing channels like generic referrals to expand Instagram campaign budgets.
- **Mitigate September Seasonal Volatility:** The third quarter shows a repeating revenue dip, peaking in September (29 orders in 2023 and 28 in 2024). Introducing seasonal promotions in late August can stabilize inventory flow and maintain baseline transaction volumes.
- **Rationalize Discount Coupons:** Descriptive analysis shows a negligible 1.4% change in average order values between coupon and non-coupon transactions. The coupon structure should be re-evaluated to target high-tier buyers instead of offering baseline discounts.